

# INTERNSHIP

## Cybersecurity Project Assignment

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|                      |                                       |
|----------------------|---------------------------------------|
| Intern ID            | NGS-CS26-KM23                         |
| Project Name         | Keylogger (Keystroke Monitoring Tool) |
| Assigned Mentor Name |                                       |

### Project Structure

```
Keylogger_KM23.zip
|
+-- Keylogger/
|   |-- keylogger.py           # core: capture, window log, rotation, encryption
|   |-- decrypt_log.py        # utility to decrypt an encrypted log
|   |-- requirements.txt       # Python dependencies
|   |-- README.md             # setup & usage guide
|   |-- REPORT.md             # architecture + detection & defense report
|   |-- .gitignore            # keeps captured data & keys out of git
|   |-- docs/
|       |-- screenshots/      # demo images
|-- logs/                     # generated at runtime
|   |-- keystrokes.log         # the captured log
|   |-- secret.key            # Fernet key (only when --encrypt is used)
```

# Installation

## System Requirements

- Python 3.12 or later
- pip (Python Package Manager)
- Internet connection (for installing dependencies)
- Windows 10/11, macOS 12+, or Linux

After downloading or cloning the project, open a terminal (Command Prompt / PowerShell on Windows) and navigate to the project folder:

```
cd Keylogger
```

## Windows Installation

**Step 1: Install Python** — download from <https://www.python.org/downloads/> and check “**Add Python to PATH**” during installation.

### Step 2: Verify Python & pip

```
python --version
pip --version
```

### Step 3: (Recommended) Create a virtual environment

```
python -m venv venv
venv\Scripts\activate
```

If PowerShell blocks activation, run:

```
Set-ExecutionPolicy -Scope Process RemoteSigned
```

### Step 4: Install required packages

```
pip install -r requirements.txt
```

You should see packages such as: **pynput**, **cryptography**.

### Step 5: Run the application

```
python keylogger.py
```

## macOS Installation

```
python3 --version          # or: brew install python
python3 -m venv venv
source venv/bin/activate
pip install -r requirements.txt
python3 keylogger.py
```

On macOS, grant your terminal **Accessibility** and **Input Monitoring** permission in System Settings → Privacy & Security.

## Linux Installation (Ubuntu/Debian)

```
sudo apt update
sudo apt install python3 python3-pip python3-venv xdotool -y
python3 -m venv venv
source venv/bin/activate
```

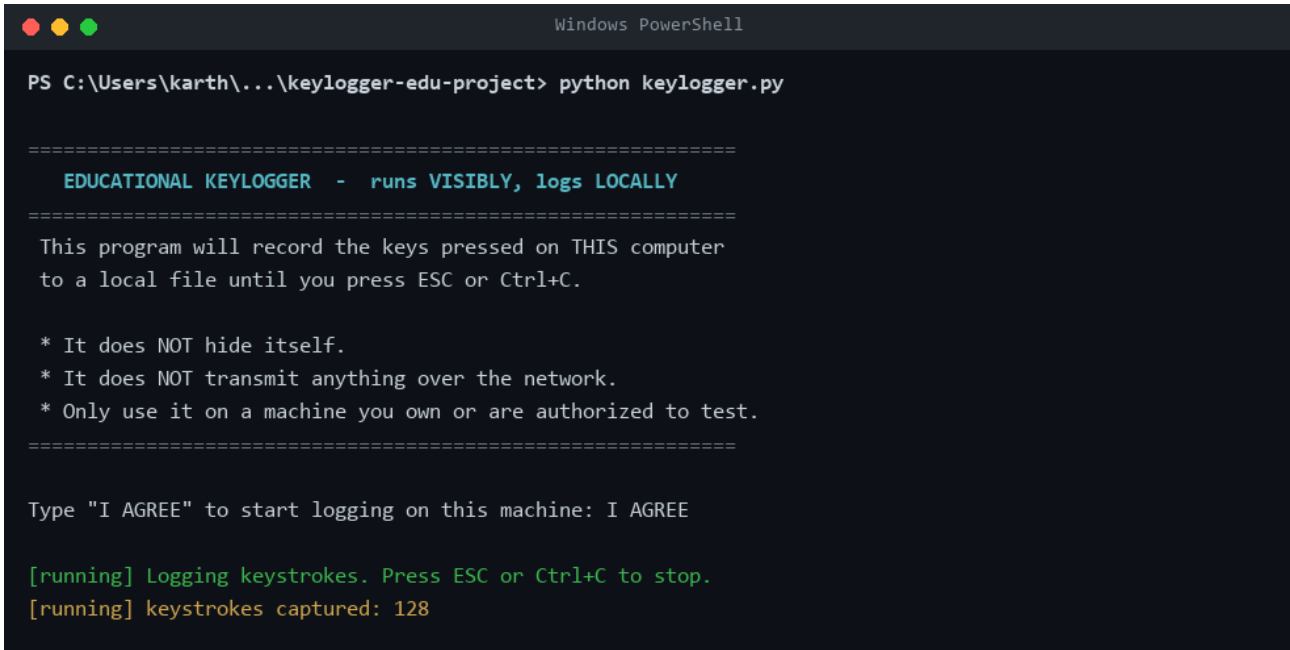
```
pip install -r requirements.txt
python3 keylogger.py
```

## Optional Flags

```
python keylogger.py --encrypt      # encrypt the log at rest
python keylogger.py --no-window   # do not record active-window titles
python decrypt_log.py             # decrypt an encrypted log
```

# Screenshots

## Application Running (visible consent banner + live counter)



```
Windows PowerShell

PS C:\Users\karth\...\keylogger-edu-project> python keylogger.py

=====
  EDUCATIONAL KEYLOGGER - runs VISIBLY, logs LOCALLY
=====

This program will record the keys pressed on THIS computer
to a local file until you press ESC or Ctrl+C.

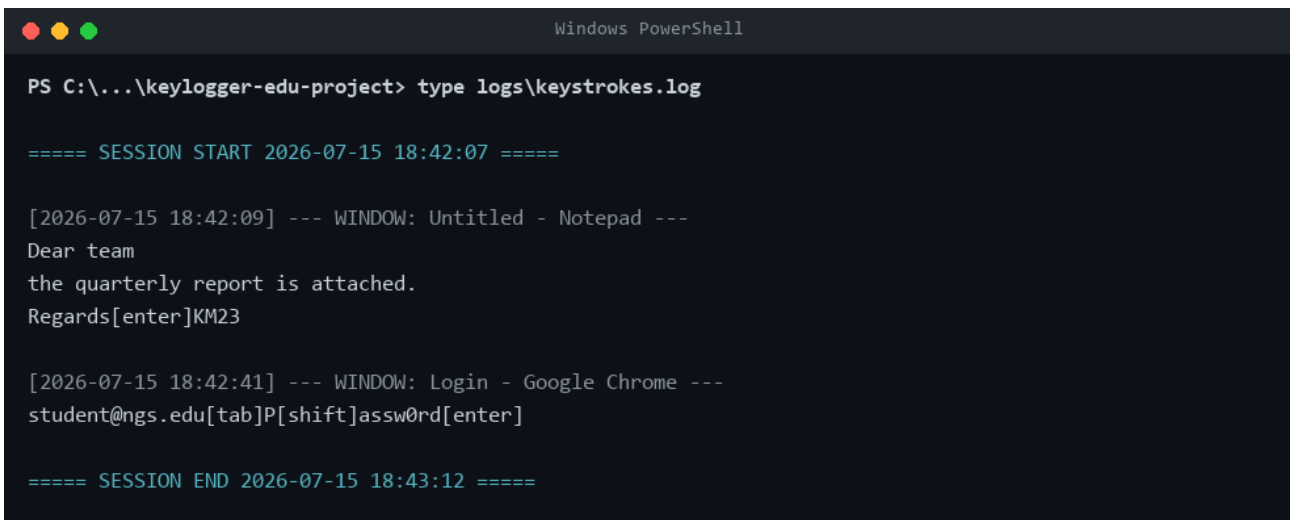
* It does NOT hide itself.
* It does NOT transmit anything over the network.
* Only use it on a machine you own or are authorized to test.
=====

Type "I AGREE" to start logging on this machine: I AGREE

[running] Logging keystrokes. Press ESC or Ctrl+C to stop.
[running] keystrokes captured: 128
```

Figure 1 — The keylogger runs visibly and requires consent.

## Captured Log File (with active-window context)



```
Windows PowerShell

PS C:\...\keylogger-edu-project> type logs\keystrokes.log

===== SESSION START 2026-07-15 18:42:07 =====

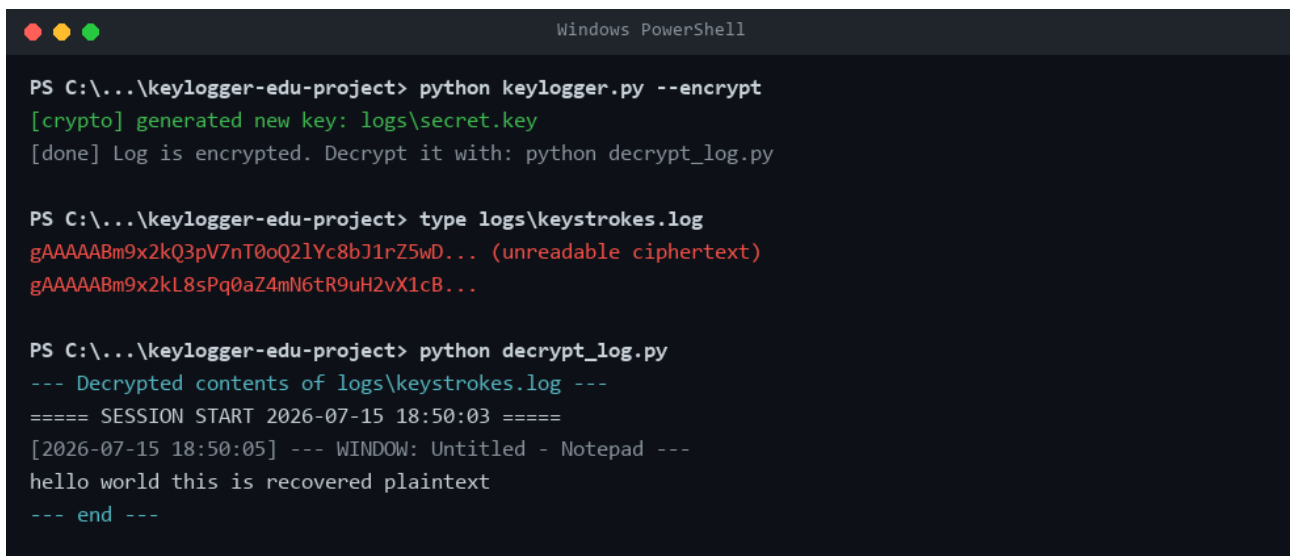
[2026-07-15 18:42:09] --- WINDOW: Untitled - Notepad ---
Dear team
the quarterly report is attached.
Regards[enter]KM23

[2026-07-15 18:42:41] --- WINDOW: Login - Google Chrome ---
student@ngs.edu[tab]P[shift]assw0rd[enter]

===== SESSION END 2026-07-15 18:43:12 =====
```

Figure 2 — Keystrokes saved locally with timestamps and window markers.

# Encrypted Log & Decryption



```
Windows PowerShell

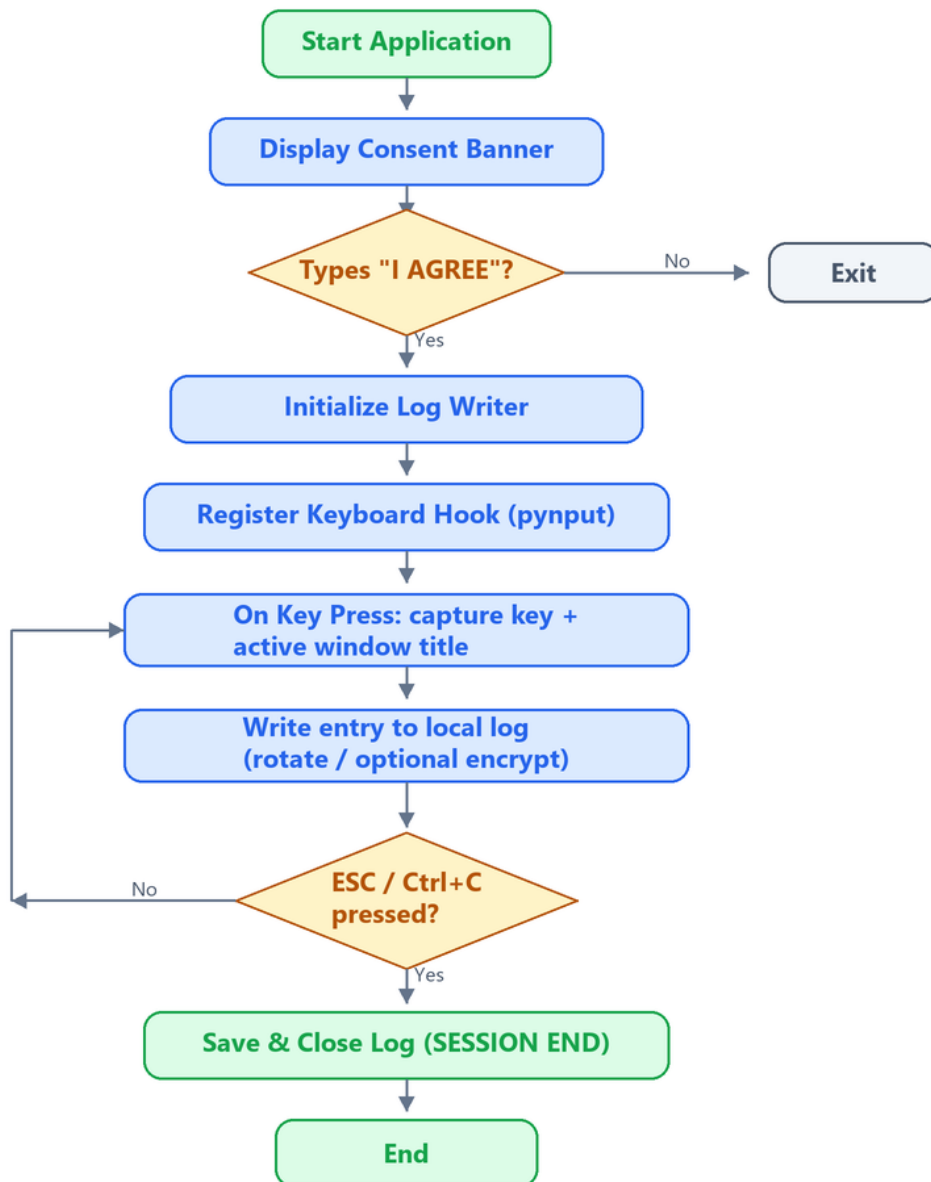
PS C:\...\keylogger-edu-project> python keylogger.py --encrypt
[crypto] generated new key: logs\secret.key
[done] Log is encrypted. Decrypt it with: python decrypt_log.py

PS C:\...\keylogger-edu-project> type logs\keystrokes.log
gAAAAABm9x2kQ3pV7nT0oQ2lYc8bJ1rZ5wD... (unreadable ciphertext)
gAAAAABm9x2kL8sPq0aZ4mN6tR9uH2vX1cB...

PS C:\...\keylogger-edu-project> python decrypt_log.py
--- Decrypted contents of logs\keystrokes.log ---
===== SESSION START 2026-07-15 18:50:03 =====
[2026-07-15 18:50:05] --- WINDOW: Untitled - Notepad ---
hello world this is recovered plaintext
--- end ---
```

Figure 3 — With --encrypt the log is stored as ciphertext and recovered via decrypt\_log.py.

# Project Workflow



# Project Description

**Keylogger (Keystroke Monitoring Tool)** is a Python-based application developed to demonstrate how keystroke capture works using an operating-system keyboard hook (via the **pynput** library). It records key presses to a local, timestamped log and tags each entry with the active window title for context. The tool is built responsibly for educational use: it runs **visibly**, requires the operator's **consent**, stores data **only on the local machine** with optional **Fernet encryption**, and performs no network transmission. This project demonstrates the practical application of input monitoring, cryptography, and cybersecurity concepts, and — through its accompanying report — how such tools are **detected and defended against**.

## Ethical & Legal Note

This tool must only be run on a computer you own or are explicitly authorized to test. Recording another person's keystrokes without informed consent is illegal under wiretapping and computer-misuse laws. The design is deliberately non-stealth, non-persistent, and local-only to keep the demonstration lawful and safe.

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## Thank You

Intern ID: NGS-CS26-KM23